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A Multimodal Approach To Detect User's Emotion

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Abstract

Emotion is a feeling, a behaviour, a state of mind of a person that is caused due to internal or external factors. Among the numerous emotions, we have considered a few emotions like happy, sad, fear, anger, surprise, love, hate, exhaust and disgust in the paper. Emotions are detected predominantly using analysis of textual contents even though there are other means such as analysis of speech, facial expressions etc. In this paper, we present a multimodal approach to detect emotions by analysing elements like the dynamics of the user's keystroke, semantics of the text and variations of the heart rate. Results of our experiments have shown that consideration of these emotion elements is found to be effective in detecting user's emotions.

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1. Introduction

Emotions are a state of feeling that results in physiological and psychological changes that influence our behavior. It can motivate us to take action. They help us to survive, thrive and avoid danger and also allow other people to understand [18]. Machines, like computers are incapable of detecting the emotion of the user. They cannot react to the user's emotional states unlike human beings. If the computers were capable of extracting the emotional state an user is going through at a particular time frame, they would have make them smarter and beneficial to many applications such as perceptual computing. It is an application of artificial intelligence in which computers interact with users and accordingly adapt to the context.

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In our system, computer detects the emotion of the user based on three elements or factors. They are: user style of typing, text content analysis and the variation in the heart rate. In user style of typing, unique parameters such as Session time, Latency, Held time, Pause rate, Capitalization rate, Typing speed and Frequency of errors [1] are considered. In the Text content analysis, two approaches are used. First approach is Java API for WordNet Searching (JAWS) and the second is using Natural Language ToolKit NLTK WordNet to determine the emotion and its intensity. Both these approaches are tested on several data sets, and NLTK approach is found to be more accurate.

In this paper, we propose to employ the above mentioned methods on text and analyse the accuracy of the algorithms to detect emotions. We detect emotion of users as normal phrases and emoticons (textual smiley's) from reviews or opinionated texts. An example of one such opinionated text is as follows: *“Well I will review it as one of the most exciting phones to ever come out. Well I do think it might be behind the times compared to older phones from Nokia like the N95 etc., but it is still a nice phone.”*

While the user types the above sentence on his computer, the heart rate is determined by taking varying skin tone as the input from webcam. The Photoplethysmography (PPG) signal is extracted from the skin pixels which are used to compute the instantaneous heart rate. Based on the variations in the heart rate, emotion is detected. Using a standard webcam, frames are captured at a particular rate (fps – frames per second) and passed through the face detection phase which acts as the input.

The remainder of this paper is organized as follows: In Section 2, a brief description of related work is presented. Then in Section 3, Methodology is discussed. In Section 4, Experiment and Results are discussed and finally, Conclusion in Section 5.

2. Related Works

There are no existing implementations for detecting emotion from Keystroke Dynamics. However several features have been proposed by Pragya Shukla and Rinky Solanki [1]. The hypothesis for emotion detection from these features is described in A. Koakowska [5].

Problem of emotion recognition from text can be formulated as follows: Let E be the set of all emotions, A be the set of all authors, and let T be the set of all possible representations of emotion-expressing texts. Let r be a function to reflect emotion e of author a from text t , i.e., $r : A \times T \rightarrow E$, then the function r would be the answer to the problem. The main problem of emotion recognition systems lies in fact that, although the definitions of E and T may be straightforward, the definitions of individual element, even subsets in both sets of E and T would be rather confusing. On one side, for the set T , new elements may add in as the languages are constantly emerging. Whereas on the other side, currently there are no standard classifications of “all human emotions” due to the complex nature of human minds, and any emotion classifications can only be seen as “Labels” annotated afterwards for different purposes[7].

There are a plethora of different techniques and methodologies to accurately detect, determine or measure the heart rate. Of all these methods, ECG (Electrocardiography) is considered as the golden standard of measure [10]. It uses electrodes (12 in number), placed on different parts of the body to record electric potential generated by the heart. The appropriate application of these electrodes can be complicated and requires professionals in this field. The readings obtained are in terms of a graph, which a layman cannot translate on his/her own. Other methods of measuring cardiac pulse involve the use of thermal imaging and Doppler phenomenon, both using optical and ultrasonic or piezoelectric measurements [10]. All the above proposed methods need sophisticated devices and are not non-contact methods.

Opinion mining is a recent subdiscipline of information retrieval which is not about the topic of a document, but with the opinion it expresses [14] [11]. In literature, opinion mining is also known as sentiment analysis [6], sentiment classification [4], opinion extraction [4], affective classification, emotions [11] and affective rating [2]

We have also referred a few other literatures on sentiment analysis, emotion analysis, methodologies to detect heart rate etc. Due to page limit constraints, we have presented only a few literatures. The review of existing literature, we found that usage of elements like Keystroke Dynamics, Textual analysis and Heart rate variation is a unique attempt to detect user's emotions.

3. Methodology

We employ Keystroke dynamics, Text based detection and Heart rate variation to detect user's emotions.

3.1. Keystroke Dynamics

It aims at finding a pattern in the typing style of the user. Typing dynamics pattern vary from person to person. It also varies depending on the current state of emotion of a user. We divide the entire process into two phases namely data collection and feature extraction.

3.1.1. Data Collection Process

A web page was created with emotion inducing videos and was circulated to the users. The objective of the web page is to record the interaction of the users with system and ascertain their emotions. Interactions of the users via keys of the system were recorded as data and stored in the database for experimentation.

3.1.2. Feature Extraction Process

An important phase is to extract the relevant features for emotion detection. We use following typing style attribute of users to capture emotions.

- **Session Time:** It is the total time spent by the user on the system. It is computed using Eq.1

$$\text{Session Time} = \text{Submit Time} - \text{User Response Time} \quad (1)$$

- **Latency:** It is the average time between two key presses in a word. It is estimated using Eq. 2

$$\text{Latency} = \frac{\sum \text{Releasing Time Of Key 1} - \sum \text{Depressing Time Of Key 2}}{\text{Charactres Per Sentence}} \quad (2)$$

- **Held Time:** It is the average time taken between a key press and a key release.
- **Pause rate:** It is the time the user spends responding to a question. It is estimated using Eq. 3

$$\text{Pause rate} = \text{User Response Time} - \text{Question Time} \quad (3)$$

- **Capitalization Rate:** It is the rate of using capital letters in a sentence which is estimated using Eq. 4

$$\text{Capitalization Rate} = \frac{\text{Capital Characters Per Sentence}}{\text{Charactres Per Sentence}} \quad (4)$$

- **Typing Speed:** It is the total number of keystrokes or words per minute.
- **Frequency of Error:** It is the use of backspace key. This rate may provide evidence of a state of confusion, as individuals who tend to delete are more likely to be confused. We estimate it using Eq. 5

$$\text{Frequency Of Error} = \frac{\sum \text{Use Of Backspace Keys}}{\text{Charactres Per Sentence}} \quad (5)$$

All the above features are recorded in the database. This is repeated for different emotions for 'n' subjects, until the values are sufficient for system calibration. Once the system is calibrated, it is ready for the users. When any user starts typing, features are extracted and most suitable emotion for those set of values are extracted from database.

3.2. Text Based Detection

We undertake two process namely Data collection and Analysis of textual documents from JAWS and NLTK approach.

3.2.1. Data Collection Process

The data sets were collected from different sources. One of them was by creating the data sets using NLTK. Using the command 'similar' (keyword), a random sentence which contains the keyword is retrieved from the NLTK corpus. The other data set used was to test the intensity of emotions. 200 pre-classified test samples which contained user reviews on products. A standard classified dataset, ISEAR [12], containing 7666 sentences, was also used for the analysis. In addition to this, the experiment was also conducted on 200 pre-classified test samples containing only positive and negative product reviews.

3.2.2. JAWS approach

The messages from the web page are first Tokenized. In lexical analysis, tokenization is the process of breaking a stream of text up into words, phrases, symbols, or other meaningful elements called tokens. These tokenized words are subjected to Parts of Speech tagging. Part of speech tagging (POS tagging or POST), also called grammatical tagging or word-category disambiguation, is the process of marking up a word in a text (corpus) as corresponding to a particular part of speech, based on both its definition, as well as its context.

Then, all the words are stemmed to obtain the corresponding root words. Stemming is the term used in linguistic morphology and information retrieval to describe the process for reducing inflected (or sometimes derived) words to their word stem, base or root form generally a written word form. A dictionary is maintained which contains some commonly used words. These words are the deciding factors of the emotion. If any new word occurs which is not present in the dictionary, then that word is searched in the WordNet dictionary which is connected through 'Java API for WordNet Searching (JAWS)'. It produces synsets of that particular word and those synsets are searched in our dictionary to match the corresponding mood. An array is maintained to store the intensity and the mood of each word in the sentence. The corresponding mood of the word is identified and its intensity is updated in the array. Finally, the mood is displayed with its intensity.

3.2.3. NLTK approach

We divide the whole text analysis into two parts namely intensity determination and emotion detection. Emotion detection step begins with pre-processing of raw text. Pre-processing is carried out both for training and testing phase. Initially Emotion laden words belonging to each category of emotions are collected from website [7]. Synsets are created for these words using WordNet tool available in NLTK python module [3]. For each word in the list stem word is determined and a new dictionary is produced.

For testing process, we input single or multiple sentences. Algorithm will remove stop words, tokenizes the sentence, consults WordNet to generate synset of these tokenized words and generates the stem (root) of each word in synset and builds a new dictionary. Match the number of matching synsets with the initial dictionary. Output the number of matched words from each emotion class. The strength of emotion belonging to a particular class depends on the number of words that match with the dictionary.

For intensity analysis purpose, complex sentences are broken to simple sentences and tokenized. Single characters are removed because they do not add value to sentiment analysis. The remaining sentence is tokenized to produce words and emotions dictionary. We built two new dictionaries namely booster dictionary and negative dictionary. Booster dictionary contains set of adjectives which enhances and degrades the meaning of noun. Booster contains 45 positive booster words like absolutely, amazingly, completely, considerably, deeply etc., and negative 20 booster words like almost, barely, hardly, just enough, kind of, kinda, kindof, kind-of, less. Negative dictionary contains 60 words like aint, aren't, cannot, cant, couldnt, darent, didnt, doesnt, ain't, aren't, can't, which can negate the meaning. In a loop, a set of trigram are processed for each of the emotion laden word in the sentiment dictionary. We use VADER-sentiment-lexicon.txt [17] to obtain the initial valence of the emotion-laden word (ELW).

ELW is searched in sentiment dictionary. In a loop, three prefix words (W) or trigrams are i.e., [W-3] [W-2] [W-1] [ELW] are searched in booster and negative dictionary.

3.3. Heart Rate Variation

Determination of emotion based on heart rate relies on the fact that the amount of light being absorbed (light that illuminates the face) depends on the volume of blood present in the region of interest (we have taken face as the ROI). The following subsections describes the process involved in doing so.

3.3.1. Data Collection Process

Handpicked images from a databases, GAPED [16] of 730 pictures and EmoMardrid [15], were used to visually elicit emotions and the raw data was extracted using webcam for further processing.

3.3.2. Extraction of PPG signal to get the Heart Rate

Video of a person are captured and the frames are collected. Each frame captured by the webcam is subjected to various stages of pre-processing such as image filtering or refining to remove unwanted facial occlusion. Skin pixels containing Photoplethysmography (PPG) information that can be used to deduce the instantaneous heart rate (iHR) are extracted.

The pixel data collected falls in the wavelength range of 540 nm to 577 nm, which corresponds to the green and yellow wavelength. It is known the haemoglobin (oxy and deoxy) absorbs light in this wavelength range [8]. Based on the amount of green intensity that is reflected (in essence absorbed), we calculate the heart rate and ultimately the emotion. The images for the slideshow were meticulously selected from the standard sets of images taken from EmoMardrid sponsored by CEACO (Cerebro, Afecto y Cognicion) and GAPED (Geneva Affective Picture Database). After capturing, each frame is subject to face detection using the Viola-Jones facial detection algorithm using Haar Cascade Classifier method [8]. This gives the contours of the face. Once the face has been detected, the Region of Interest is set to the boundaries of the face and is subjected to image filtering to remove unwanted noise. The face detected and the videos frames are first converted from RGB color space to YCbCr color space and threshold values for each of Y, Cb and Cr are set to detect the skin (face of the subject under study). The Y, indicating the intensity and Cb, Cr respectively denotes the blue and the red chromatic color components.

The optimal values of these different parameters were set after experimentation. Simultaneously, the frames are also converted from RGB to CIE $L^*u^*v^*$ color spaces. The advantages of using CIE LUV are that it maps the color spaces into the human visual spectrum and it is device independent. Conversions from one color space to another were performed using OpenCV's [13] built-in functions (`cv2.cvtColor`). Since the color components represented by u^* falls in the required wavelength (green) this is extracted and others are filtered out.

After filtering out the unwanted data, the filtered image (with reduced noise) is converted from the time domain to frequency domain using Fast Fourier Transforms (FFT). These frequencies are mapped to specific points called frequency bins using which the heart rate (in beats per minute) at that instant can be obtained.

3.3.3. Translating HRV to Emotion State

A slideshow of images taken from the standard image database are used to elicit emotion. For each of these images, 300 samples are taken at various time intervals, which denote the heart rate at that instant. Each image is made to stand for about 16 seconds where the images are grouped as positive, negative and neutral images. The neutral images are used as a dummy, to get the heart rate back to its original resting state after each elicitation (either positive or negative) [9]. The iHR generated for each of these images are stored in the SQLite Database for further analysis.

The database houses a table containing the iHR for each of the images displayed corresponding to the user id, filename, and at various instances. In addition to this, a table containing the difference or the variation of the heart rate from neutral to an elicited state of mind is stored.

The average value of the iHR generated for each image is computed. A slideshow of images (using OpenCV [13]) are displayed to the subject under evaluation. The average iHR obtained during slideshow, of the images used to elicit either positive or negative emotions, are subtracted from the immediately preceding neutral set of images. The neutral images act as a baseline for the computation of the Heart Rate Variation. The amount by which the heart rate varies i.e., difference of the average heart rate when neutral images were being shown and the average heart rate

when the subject was elicited is computed and stored in the database. These values are computed for each individual during the training phase and are used as standard means for detecting the emotions during testing.

4. Experiments and Results

All the three modules were subjected to tests for different datasets and the results are as follows.

4.1. Keystroke Dynamics

From the 100 data samples manually collected, following hypotheses are drawn:

Using this method, the two extremes of emotions, happy and sad can be deduced, which respectively assesses the

- Frequency of error is more, if the user is sad. This is found to be true in 70% of the cases.
- Typing speed is more, if the user is happy and found to be true in 60% of the cases.
- Capitalization rate is found to be more, if the user is happy in 80% of the cases.
- No regular trend was found in pause rate.
- Held time is found to be more, if the user is happy in 50% of the cases.
- Latency is found to be more, if the user is happy in 50% of the cases.
- Session time is found to be more, if the user is sad in 80% of the cases.

Using this method, the two extremes of emotions, happy and sad can be deduced, which respectively assesses the emotion state of the individual to be either positive or negative. From the above parameters, Frequency of Error, Capitalization Rate and the Session Time provided a greater degree of accuracy in detection of emotion. An average accuracy with respect to the prediction of the all these emotion was found to be 77.67%.

4.2. Text Based Detection

Both the approaches were tested for several cases of emotion categories. The results are as follows:

4.2.1. Using JAWS

From Table 1, Happy, Fear and Surprise have greater success rate. Hence they can be used to efficiently determine emotions. We have considered the five emotions as they are considered to be basic state of emotions [12] and also the datasets pertaining to these emotions are abundant. Results obtained using this method is provided in Table 1. An average accuracy with respect to the prediction of the all these emotion was found to be 77.4%.

Table 1. Accuracy of JAWS Approach

SL. No	Emotion	No. of test cases [12]	Accurate Prediction (Nos.)	Success Rate (%)
1	Happy	100	82	82
2	Sad	100	64	64
3	Fear	100	84	84
4	Surprise	100	90	90
5	Anger	100	67	67

4.2.2. Using NLTK

We have considered the basic five emotions as in JAWS to obtain accuracy of 92%, 73%, 86%, 91% and 100%. We have also considered additional emotion factors like fear, surprise, exhaust and disgust for capturing user's emotions. As shown in the Table 2, emotion elements like Hate, Happy, Fear, Surprise and Exhaust have greater success rate. Average accuracy of determining the emotions using this method was found to be 88.7%.

Table 2. Accuracy of NLTK Approach

SL. No	Emotion	No. of test cases [12]	Accurate Prediction (Nos.)	Success Rate (%)
1	Happy	100	92	92
2	Sad	100	73	73
3	Love	100	86	86
4	Anger	100	91	91
5	Hate	100	100	100
6	Fear	100	100	100
7	Surprise	100	95	95
8	Exhaust	100	93	93
9	Disgust	100	69	69

4.3. Heart Rate Variation

4.3.1. Testing with different types of inputs

- **Occlusion:** If any objects are blocking the face, then the algorithm filters out these occluded regions of the face and tries to extract the PPG information from the remaining portions of the face.
- **Side View:** The algorithm fails to detect the face, which is the key for further processing when the subject is not viewing the camera or the display directly. Since the face cannot be detected under these conditions, the heart rate and further the emotion state of the person cannot be deduced.
- **Low light:** During low light condition, the face recognition and other filtering techniques worked without any glitch. But the accuracy of the heart rate variation in these conditions dropped.

4.3.2. Standard Testing with many samples

a

uid	file_name	emo_state	diff
Filter	Images/Testing/P026.jpg	Filter	Filter
1	Images/Testing/P026.jpg	Positive	15.6792194...
1	Images/Testing/P026.jpg	Positive	8.78838168...
1	Images/Testing/P026.jpg	Positive	3.15992173...
2	Images/Testing/P026.jpg	Negative	1.70793942...
3	Images/Testing/P026.jpg	Negative	4.31167987...
4	Images/Testing/P026.jpg	Positive	5.08088724...
4	Images/Testing/P026.jpg	Positive	7.46634299...

b

uid	file_name	emo_state	diff
Filter	Images/Testing/q.jpg	Filter	Filter
2	Images/Testing/q.jpg	Negative	0.75125332...
3	Images/Testing/q.jpg	Negative	4.35823992...
4	Images/Testing/q.jpg	Negative	0.65719664...
4	Images/Testing/q.jpg	Positive	3.51901344...

Fig. 1. (a) dB Picture Positive; (b) dB Picture Negative.

Experiments were conducted with 100 samples to analyse and determine the accuracy of the proposed system. Figure 1 (a) and (b) shows a subset of training and test sample that were collected and used. We obtained an accuracy of 71% for positive emotions and 75% for negative emotions.

5. Conclusion

We have discussed an approach to detect emotions of users by analysing elements like the dynamics of the user's keystroke, semantics of the text and variations of the heart rate. Currently, many applications use only textual information to detect user's emotions. From the literature, accuracies of a few well known approaches to detect emotions from textual information on different domains vary between 70 % and 95 %. We believe textual information provides only a fraction of user's emotions and incomplete emotions in some cases, like in the where

the users are distressed, anxious etc. By analysing the information obtained by keystroke, heart rate in addition to textual information, effective emotions of users can be ascertained. Our approach provides an average accuracies of 77.67%, 88.7% and 73% with keystrokes, textual and heart rate information respectively on different datasets.

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